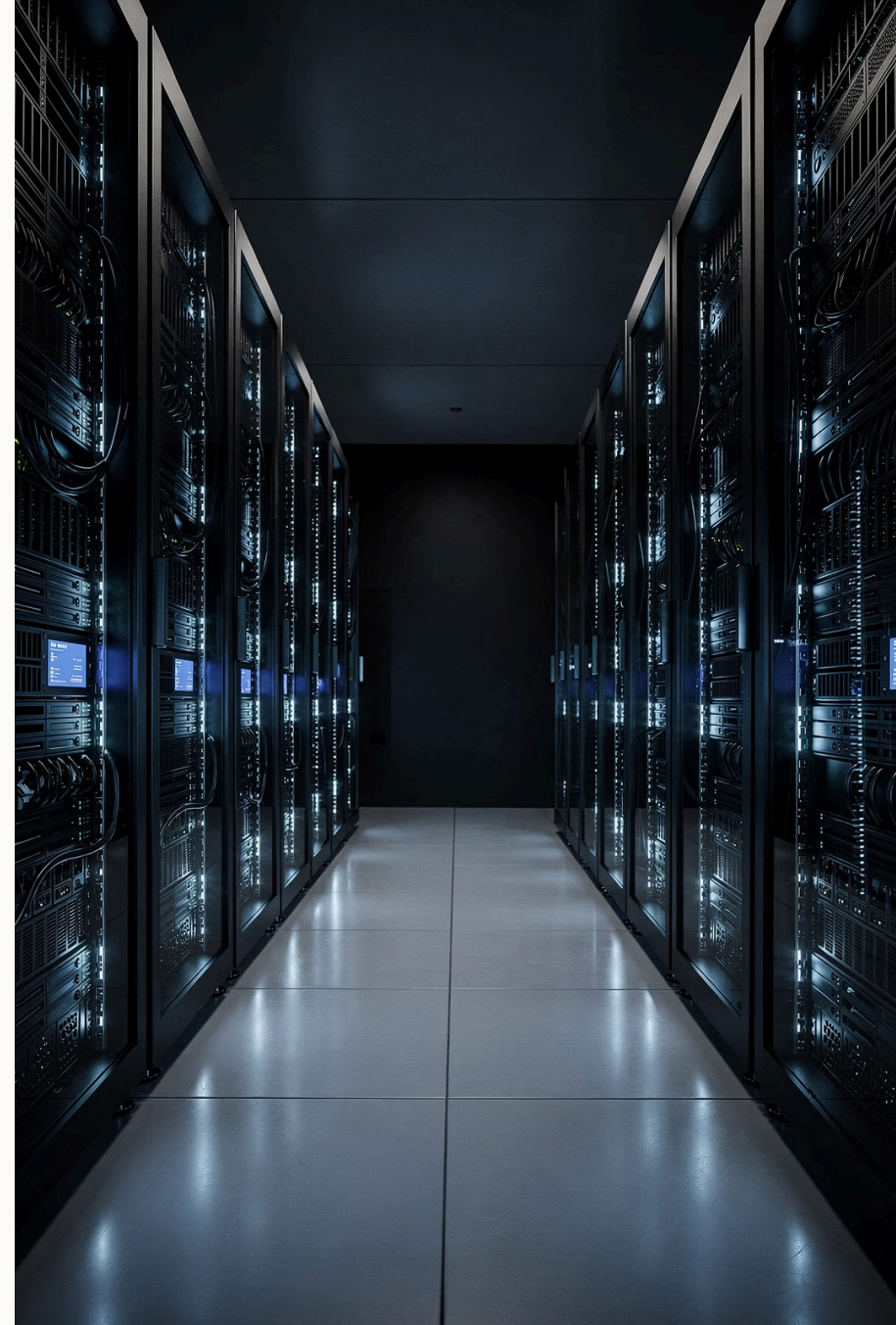


Database, DBMS, RDBMS, SQL, MySQL & SQL*Plus

Unit I · Lecture 2 · MySQL (SQL/PL-SQL)

Presented by Renuka Parmar · BCA Program





Quick Recap: Lecture 1

Last session, we laid the foundation. Here's what we covered:

Why SQL?

The universal language for data management

Career Scope

DBA, Data Analyst, Backend Developer

History of SQL

From IBM's SEQUEL to ANSI standard

Course Roadmap

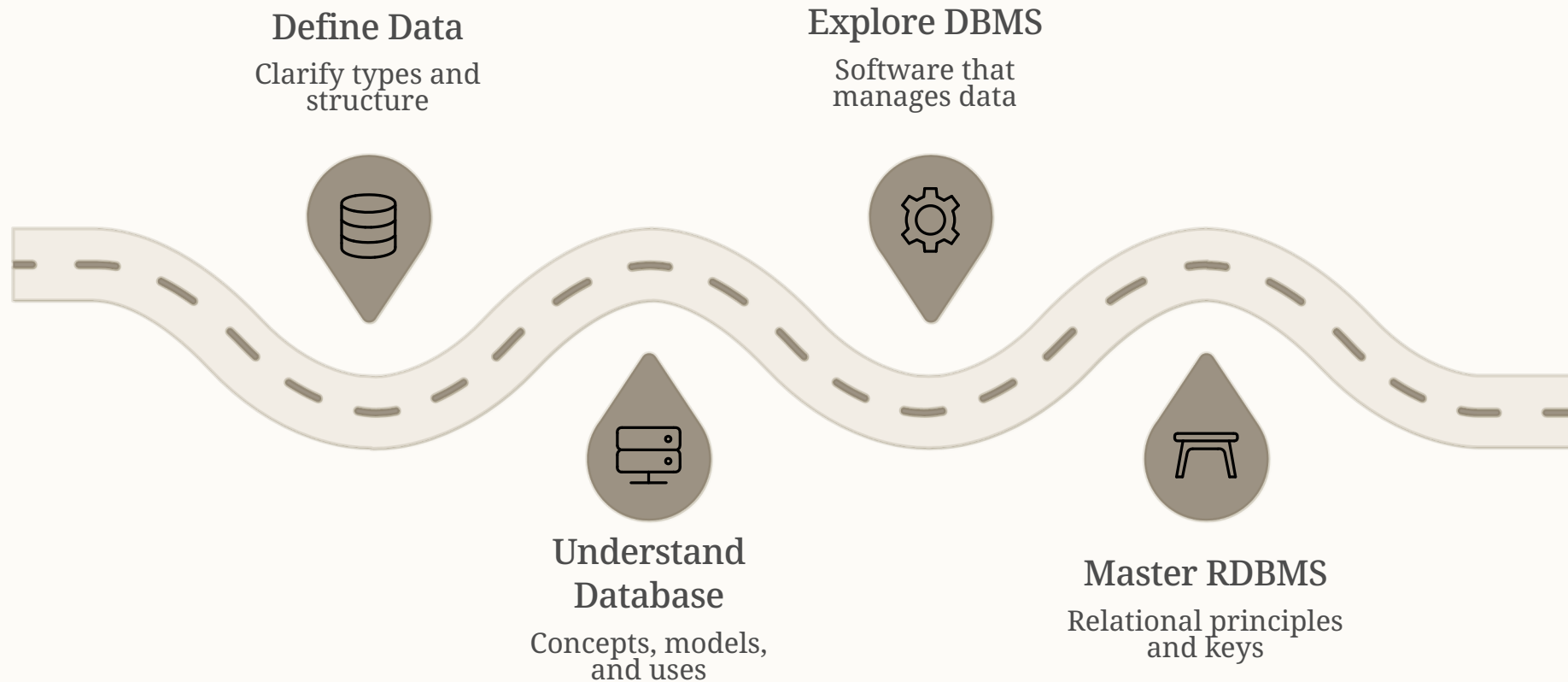
DDL, DML, DQL, DCL, TCL & PL/SQL



Today: We explore the technologies that make SQL possible — Data → Database → DBMS → RDBMS → SQL → MySQL.

Today's Learning Objectives

By the end of this lecture, you will be able to:



Each concept builds on the previous one — master them in sequence for a complete picture of the database ecosystem.



CONCEPT 1

What is Data?

Data is a collection of raw, unprocessed facts, figures, or symbols that have no meaning on their own.

Types: Structured (tables), Semi-structured (JSON/XML), Unstructured (images, emails)

Raw Data

Student marks: 78, 92, 65

Processed Info

Average score: 78.3% — Pass

CONCEPT 2

What is a Database?

A **Database** is an organized, structured collection of interrelated data stored electronically for easy access, retrieval, and management.

- **Library:** Books catalogued by title, author, genre
- **Bank:** Customer accounts, transactions, balances
- **University:** Student records, courses, faculty details



A database is NOT just a folder of files — it enforces structure, relationships, and rules.



Characteristics of a Database



Organized Storage

Data is arranged in a structured format — tables, rows, and columns — for efficient access.



Data Sharing

Multiple users or applications can access the same data simultaneously.



Security

Access controls and authentication protect sensitive data from unauthorized use.



Backup & Recovery

Built-in mechanisms ensure data is not permanently lost in case of failure.



Data Consistency

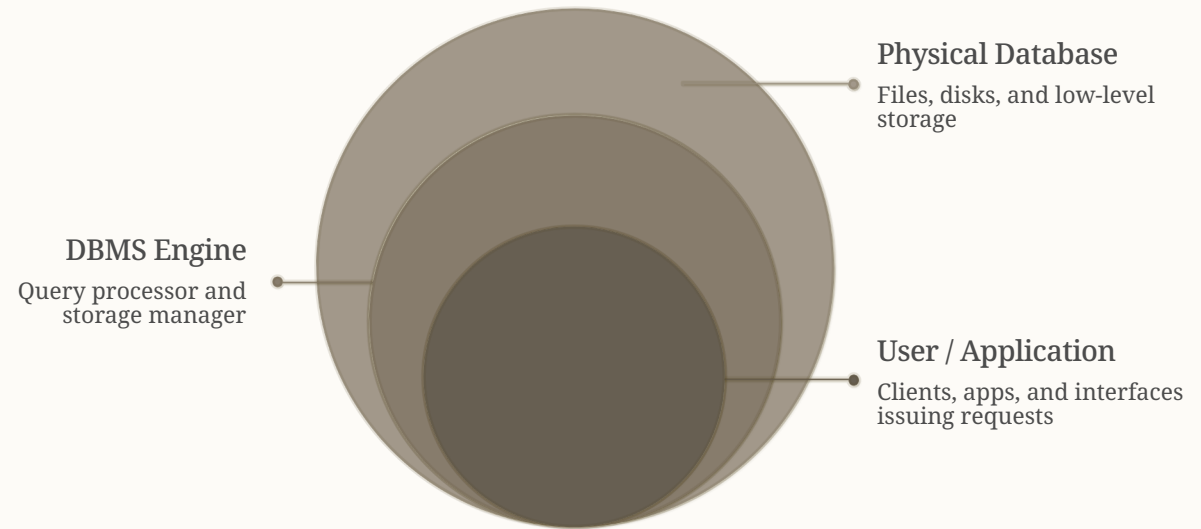
Rules (constraints) ensure that data remains accurate and free from contradictions.

What is a DBMS?

A **Database Management System (DBMS)** is software that acts as an interface between users/applications and the database. It manages how data is stored, organized, retrieved, and secured.

- **Purpose:** Eliminate data redundancy and provide controlled access
- **Examples:** MS Access, dBase, FoxPro, FileMaker

"Think of DBMS as the librarian who manages all books in a library."



Functions of DBMS



Data Storage

Organizes and stores data efficiently on physical media using file structures.



Data Retrieval

Allows users to query data using commands (e.g., SQL SELECT statements).



Data Security

Grants or restricts access using user roles, passwords, and permissions.



Backup & Recovery

Automatically backs up data and restores it after crashes or failures.



Multi-user Access

Manages concurrent access by multiple users without data conflicts.

Advantages vs. Limitations of DBMS

✓ Advantages	⚠ Limitations
Reduces data redundancy and duplication	High cost of hardware and software setup
Centralized data management and control	Complex to design and maintain
Improved data security and privacy	Requires trained database administrators (DBAs)
Easy data sharing across departments	System failure can affect all users at once
Data integrity through constraint enforcement	Performance may degrade with very large datasets



CONCEPT 4

What is an RDBMS?

A Relational DBMS (RDBMS)
stores data in **tables** (relations)
and establishes **relationships**
between them using keys.

Examples: MySQL, Oracle, PostgreSQL, SQL Server, DB2

Table

2D grid of rows & columns

Row (Tuple)

One complete record

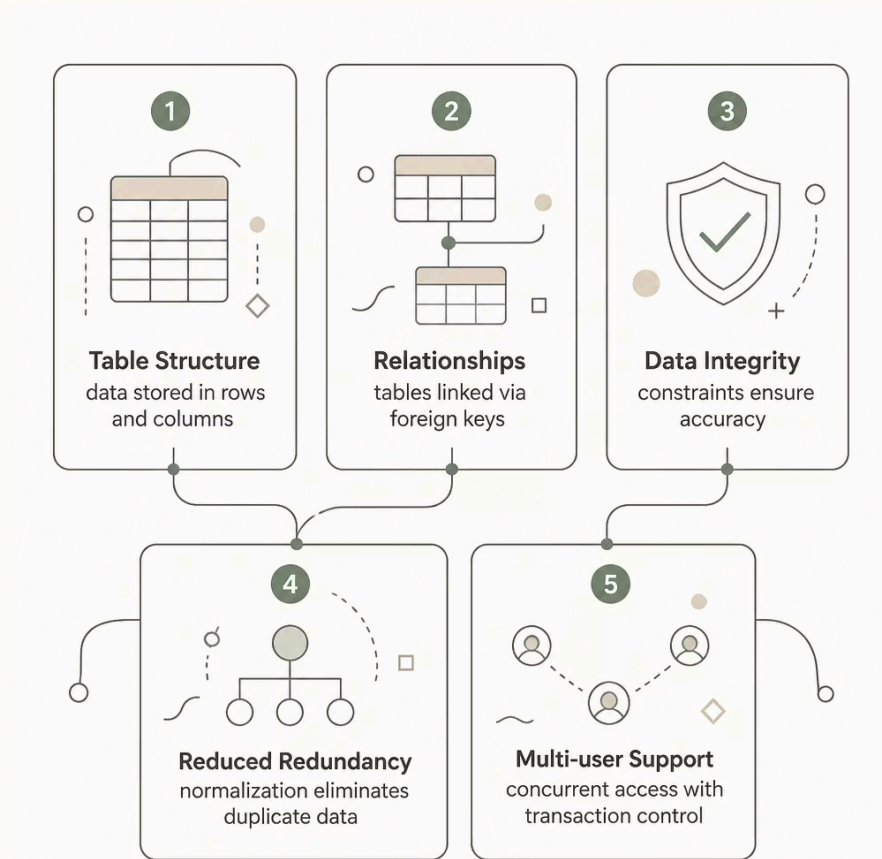
Column
(Attribute)

A specific data field

Primary Key

Unique identifier per row

Features of RDBMS



RDBMS is the backbone of modern data-driven applications. Its structured approach ensures reliability, scalability, and security.

- Used by **Google, Facebook, Amazon, Banks**
- Supports **ACID transactions** (Atomicity, Consistency, Isolation, Durability)
- Enables **JOIN operations** across related tables

DBMS vs. RDBMS

Feature	DBMS	RDBMS
Storage	Flat files or hierarchical structures	Organized tables with rows & columns
Relationships	No formal relationship between data	Tables linked via primary/foreign keys
Normalization	Not supported	Fully supported to reduce redundancy
Redundancy	High — data may be duplicated	Low — normalization minimizes duplication
Security	Limited access control	Role-based, multi-level security
Examples	MS Access, FoxPro, dBase	MySQL, Oracle, PostgreSQL, SQL Server

What is SQL?

Structured Query Language (SQL)

is the standard language used to communicate with relational databases. It allows users to create, read, update, and delete data.

- Developed by IBM in the 1970s; became ANSI standard in 1986
- Used by virtually every RDBMS: MySQL, Oracle, PostgreSQL
- Not a programming language — it's a **query language**



SQL is to databases what English is to communication — a universal standard.



Features of SQL



Easy to Learn

English-like syntax makes it beginner-friendly



ANSI Standard

Recognized globally by all major RDBMS platforms



Portable

Write once, run on MySQL, Oracle, SQL Server



High Performance

Optimized query execution on large datasets



Secure

Supports user roles, privileges, and encryption



What is MySQL?

MySQL is a free, open-source **Relational Database Management System (RDBMS)** developed by MySQL AB (now owned by Oracle). It uses SQL as its query language.

- **Open Source:** Free to download and use
- **Widely Used:** Powers WordPress, Facebook, Twitter, YouTube
- **Cross-Platform:** Runs on Windows, Linux, macOS
- **Architecture:** Client-Server model with SQL engine



MySQL's mascot is a dolphin named **Sakila** — fun fact for your first MySQL lab!

Features of MySQL



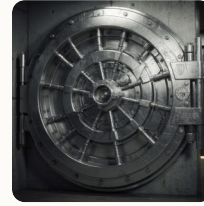
Fast & Efficient

Optimized query engine handles millions of records at high speed



Reliable

ACID-compliant transactions ensure data integrity and consistency



Secure

Password encryption, user privileges, and SSL connection support



Scalable

Scales from small apps to large enterprise-level databases



Cross-Platform

Compatible with Windows, Linux, macOS, and cloud environments



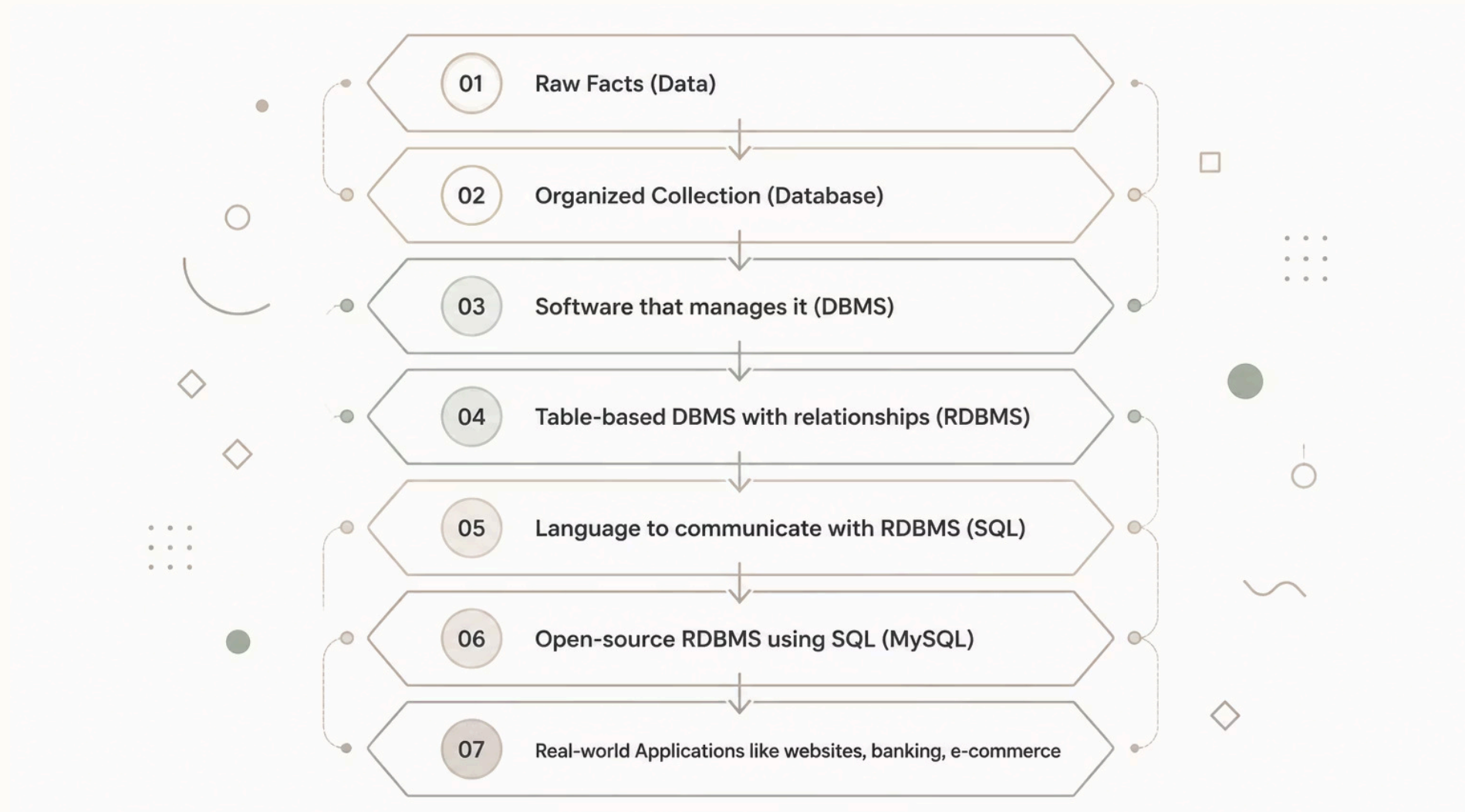
Multi-user

Supports concurrent connections from many users simultaneously

SQL vs. MySQL vs. SQL*Plus

Feature	SQL	MySQL	SQL*Plus
Type	Query Language	RDBMS Software	Command-line Tool
Developer	IBM / ANSI	Oracle Corporation	Oracle Corporation
Purpose	Standard for querying databases	Stores & manages data using SQL	Interface to run SQL on Oracle DB
Usage	Used inside all RDBMS	Web apps, enterprise software	Oracle database administration
Standalone?	No — needs an RDBMS	Yes — complete RDBMS	No — requires Oracle RDBMS

Concept Map: The Complete Picture



Every concept in today's lecture is a building block. **Data** is the raw material; **MySQL** is the engine; **SQL** is the steering wheel — together, they power every modern application.

Quiz & Assignment

Quick MCQ Quiz 🧠

1. Which of these is an RDBMS? (a) FoxPro (b) MySQL (c) MS Word (d) Excel
2. SQL stands for? (a) Simple Query Language (b) Structured Query Language (c) System Query Language (d) None
3. A row in a table is also called a? (a) Attribute (b) Column (c) Tuple (d) Key
4. Who developed MySQL? (a) IBM (b) Microsoft (c) MySQL AB / Oracle (d) Google
5. SQL*Plus is an interface for which RDBMS? (a) MySQL (b) PostgreSQL (c) Oracle (d) SQL Server



Assignment

Prepare a comparison table using **your own real-life examples** for:

- DBMS vs. RDBMS
- SQL vs. MySQL
- MySQL vs. SQL*Plus

Format: Handwritten or typed, minimum 5 rows per comparison. Submit next class.

Key Takeaways & Next Lecture

01

Data

Raw, unprocessed facts — the starting point of everything

02

Database & DBMS

Organized storage + software to manage it efficiently

03

RDBMS

Table-based DBMS with relationships, keys, and integrity

04

SQL & MySQL

SQL is the language; MySQL is the open-source RDBMS that speaks it

05

SQL*Plus

Oracle's command-line tool to execute SQL on Oracle databases

"Understanding concepts is the first step toward becoming an excellent database developer."

Next Lecture: SQL Commands — DDL · DML · DQL · DCL · TCL